LP3 Assignment B4

Roll no: 41168

Problem Statement:
Implement Gradient Descent Algorithm to find the local minima of a function. For example, find the local minima of the function $y=(x+3)^2$ starting from the point x=2.

```
import math

def gradient_descent(f, x, lr=0.01, tol=1e-9, max_iter=100_000, h=1e-6):
    for _ in range(max_iter):
        grad = (f(x + h) - f(x - h)) / (2 * h)
        step = lr * grad
        x -= step
        if abs(step) < tol:
            return x, f(x)
    return None


if __name__ == "__main__":
    expr = input("f(x) = ")                    # Format: x**2 - 4*x + 3
    x0 = float(input("Starting x = "))

    f = lambda x: eval(expr, {**vars(math), "x": x})   # math functions (sin, exp, ...) allowed
    result = gradient_descent(f, x0)
    if result:
        print(f"Local minima: x = {result[0]:.6f}, f(x) = {result[1]:.6f}")
    else:
        print("No local minima found (function may be unbounded below).")
```

OUTPUT:

```
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$ python3 LP3B4.py
f(x) = x**2 + 4*x - 10
Starting x = 5
Local minima: x = -2.000000, f(x) = -14.000000
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$ python3 LP3B4.py
f(x) = x**3 - 6*x**2 + 5*x + 15
Starting x = 3
Local minima: x = 3.527525, f(x) = 1.871549
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$ python3 LP3B4.py
f(x) = 4*x - 3
Starting x = 0
No local minima found (function may be unbounded below).
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$
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chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$
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